

CSC3034 Computational Intelligence Tutorial Performance Metrics

1. **Percent correct.** A neural network has been designed to distinguish between four classes, A, B, C, and D. The neural network has two output neurons with the sigmoid activation function. The raw values from the output neurons using the 15 sets of input data are recorded. The expected class of each input data set is provided alongside with the raw values in Table 1.

Table 1: Raw output values and expected classes

Index	Output Neuron 1	Output Neuron 2	Expected Class
1	0.623	0.028	C
2	0.112	0.225	A
3	0.809	0.228	B
4	0.349	0.284	A
5	0.664	0.359	B
6	0.810	0.304	C
7	0.673	0.818	D
8	0.465	0.439	A
9	0.275	0.972	B
10	0.788	0.829	D
11	0.624	0.658	D
12	0.890	0.359	C
13	0.699	0.478	B
14	0.674	0.984	D
15	0.788	0.366	C

To determine the predicted class of a set of input data, the raw values of the output neurons need to be translated into a boolean value using a threshold. A raw value above the threshold is translated to **True**, otherwise, to **False**. The combination of the translated outputs provides the predicted class for the set of input data as shown in Table 2.

Table 2: Predicted class based on translated outputs

Output Neuron 1	Output Neuron 2	Predicted Class
False	False	A
False	True	B
True	False	C
True	True	D

- (a) Using the threshold of **0.5**, calculate the percent correct for
- all the input data (overall percent correct)

Index	Output Neuron 1	Output Neuron 2	Predicted Class	Expected Class
1	0.623	0.028	C	C
2	0.112	0.225	A	A
3	0.809	0.228	C	B
4	0.349	0.284	A	A
5	0.664	0.359	C	B
6	0.810	0.304	C	C
7	0.673	0.818	D	D
8	0.465	0.439	A	A
9	0.275	0.972	B	B
10	0.788	0.829	D	D
11	0.624	0.658	D	D
12	0.890	0.359	C	C
13	0.699	0.478	C	B
14	0.674	0.984	D	D
15	0.788	0.366	C	C

Solution:

$$\frac{12}{15} \times 100 = 80\%$$

- the input data that are supposed to be D

Solution:

$$\frac{4}{4} \times 100 = 100\%$$

- (b) Using the threshold of **0.3**, calculate the percent correct for
- i. all the input data (overall percent correct)

Index	Output Neuron 1	Output Neuron 2	Predicted Class	Expected Class
1	0.623	0.028	C	C
2	0.112	0.225	A	A
3	0.809	0.228	C	B
4	0.349	0.284	C	A
5	0.664	0.359	D	B
6	0.810	0.304	D	C
7	0.673	0.818	D	D
8	0.465	0.439	D	A
9	0.275	0.972	B	B
10	0.788	0.829	D	D
11	0.624	0.658	D	D
12	0.890	0.359	D	C
13	0.699	0.478	D	B
14	0.674	0.984	D	D
15	0.788	0.366	D	C

Solution:

$$\frac{7}{15} \times 100 = 46.67\%$$

- ii. the input data that are supposed to be D

Solution:

$$\frac{4}{4} \times 100 = 100\%$$

2. **Errors.** Using the data in Question 1, consider **False** as 0 and **True** as 1, identify the expected output of the output neurons for each set of input data.

Index	Output Neuron 1	Output Neuron 2	Expected Class	Expected Output Neuron 1	Expected Output Neuron 2
1	0.623	0.028	C	1	0
2	0.112	0.225	A	0	0
3	0.809	0.228	B	0	1
4	0.349	0.284	A	0	0
5	0.664	0.359	B	0	1
6	0.810	0.304	C	1	0
7	0.673	0.818	D	1	1
8	0.465	0.439	A	0	0
9	0.275	0.972	B	0	1
10	0.788	0.829	D	1	1
11	0.624	0.658	D	1	1
12	0.890	0.359	C	1	0
13	0.699	0.478	B	0	1
14	0.674	0.984	D	1	1
15	0.788	0.366	C	1	0

Note that for this question we will not use the threshold to translate the raw values of the output neurons but just use them as it is.

- (a) Calculate the average sum-squared error for the system.

Solution:

$$\text{average sum-squared error, } e = \frac{\text{total error, } E_t}{\text{total number of patterns, } m}$$

$$\text{total error, } E_t = 0.5 \sum_{k=1}^m \sum_{j=1}^q (b_{kj} - y_{kj})^2$$

Solution:

No.	y_{k1}	y_{k2}	b_{k1}	b_{k2}	$e_{k1} = b_{k1} - y_{k1}$	$e_{k2} = b_{k2} - y_{k2}$	e_{k1}^2	e_{k2}^2	$e_{k1}^2 + e_{k2}^2$
1	0.623	0.028	1	0	0.377	-0.028	0.142129	0.000784	0.142913
2	0.112	0.225	0	0	-0.112	-0.225	0.012544	0.050625	0.063169
3	0.809	0.228	0	1	-0.809	0.772	0.654481	0.595984	1.250465
4	0.349	0.284	0	0	-0.349	-0.284	0.121801	0.080656	0.202457
5	0.664	0.359	0	1	-0.664	0.641	0.440896	0.410881	0.851777
6	0.81	0.304	1	0	0.19	-0.304	0.0361	0.092416	0.128516
7	0.673	0.818	1	1	0.327	0.182	0.106929	0.033124	0.140053
8	0.465	0.439	0	0	-0.465	-0.439	0.216225	0.192721	0.408946
9	0.275	0.972	0	1	-0.275	0.028	0.075625	0.000784	0.076409
10	0.788	0.829	1	1	0.212	0.171	0.044944	0.029241	0.074185
11	0.624	0.658	1	1	0.376	0.342	0.141376	0.116964	0.25834
12	0.89	0.359	1	0	0.11	-0.359	0.0121	0.128881	0.140981
13	0.699	0.478	0	1	-0.699	0.522	0.488601	0.272484	0.761085
14	0.674	0.984	1	1	0.326	0.016	0.106276	0.000256	0.106532
15	0.788	0.366	1	0	0.212	-0.366	0.044944	0.133956	0.1789
Total									≈ 4.78

$$E_t = 0.5 \times 4.78 = 2.39$$

$$e = \frac{2.39}{15} = 0.1595$$

$$\text{(optional) } e \text{ per output neuron} = \frac{e}{2} = \frac{0.1595}{2} = 0.07975$$

(b) Calculate the mean absolute error for the system.

Solution:

$$\text{mean absolute error, } E_{ma} = \frac{1}{mq} \sum_{k=1}^m \sum_{j=1}^q |b_{kj} - y_{kj}|$$

number of patterns, $m = 15$

number of output neurons, $q = 2$

No.	y_{k1}	y_{k2}	b_{k1}	b_{k2}	$e_{k1} = b_{k1} - y_{k1}$	$e_{k2} = b_{k2} - y_{k2}$	$ e_{k1} + e_{k2} $
1	0.623	0.028	1	0	0.377	-0.028	0.405
2	0.112	0.225	0	0	-0.112	-0.225	0.337
3	0.809	0.228	0	1	-0.809	0.772	1.581
4	0.349	0.284	0	0	-0.349	-0.284	0.633
5	0.664	0.359	0	1	-0.664	0.641	1.305
6	0.81	0.304	1	0	0.19	-0.304	0.494
7	0.673	0.818	1	1	0.327	0.182	0.509
8	0.465	0.439	0	0	-0.465	-0.439	0.904
9	0.275	0.972	0	1	-0.275	0.028	0.303
10	0.788	0.829	1	1	0.212	0.171	0.383
11	0.624	0.658	1	1	0.376	0.342	0.718
12	0.89	0.359	1	0	0.11	-0.359	0.469
13	0.699	0.478	0	1	-0.699	0.522	1.221
14	0.674	0.984	1	1	0.326	0.016	0.342
15	0.788	0.366	1	0	0.212	-0.366	0.578
Total							10.182

$$E_{ma} = \frac{1}{15 \times 2} (10.182) = 0.3394$$

(c) Calculate the maximum absolute error for the system.

Solution:

$$\begin{aligned} \text{maximum absolute error, } E_{\max} &= \max_{k,j} |b_{kj} - y_{kj}| \\ &= \text{maximum value of } |e_{k1}| \text{ and } |e_{k2}| \\ &= 0.809 \end{aligned}$$

(d) Calculate the normalised error for the system.

Solution:

$$\text{normalised error, } E_n = \frac{\text{total error, } E_t}{\text{mean error, } E_{\text{mean}}}$$

$$\text{total error, } E_t = 0.5 \sum_{k=1}^m \sum_{j=1}^q (b_{kj} - y_{kj})^2 = 2.39$$

$$\text{mean error, } E_{\text{mean}} = 0.5 \sum_{k=1}^m \sum_{j=1}^q (b_{kj} - \mu_j)^2$$

$$\text{mean of expected output, } \mu_j = \frac{1}{m} \sum_{k=1}^m b_{kj}$$

No.	y_{k1}	y_{k2}	b_{k1}	b_{k2}	$e_{k1} = b_{k1} - \mu_1$	$e_{k2} = b_{k2} - \mu_2$	e_{k1}^2	e_{k2}^2	$e_{k1}^2 + e_{k2}^2$
1	0.623	0.028	1	0	0.4667	-0.5333	0.2178	0.2844	0.5022
2	0.112	0.225	0	0	-0.5333	-0.5333	0.2844	0.2844	0.5688
3	0.809	0.228	0	1	-0.5333	0.4667	0.2844	0.2178	0.5022
4	0.349	0.284	0	0	-0.5333	-0.5333	0.2844	0.2844	0.5688
5	0.664	0.359	0	1	-0.5333	0.4667	0.2844	0.2178	0.5022
6	0.81	0.304	1	0	0.4667	-0.5333	0.2178	0.2844	0.5022
7	0.673	0.818	1	1	0.4667	0.4667	0.2178	0.2178	0.4356
8	0.465	0.439	0	0	-0.5333	-0.5333	0.2844	0.2844	0.5688
9	0.275	0.972	0	1	-0.5333	0.4667	0.2844	0.2178	0.5022
10	0.788	0.829	1	1	0.4667	0.4667	0.2178	0.2178	0.4356
11	0.624	0.658	1	1	0.4667	0.4667	0.2178	0.2178	0.4356
12	0.89	0.359	1	0	0.4667	-0.5333	0.2178	0.2844	0.5022
13	0.699	0.478	0	1	-0.5333	0.4667	0.2844	0.2178	0.5022
14	0.674	0.984	1	1	0.4667	0.4667	0.2178	0.2178	0.4356
15	0.788	0.366	1	0	0.4667	-0.5333	0.2178	0.2844	0.5022
Average			$\mu_1 = 0.5333$	$\mu_2 = 0.5333$	Total				≈ 7.466

$$\text{mean error, } E_{\text{mean}} = 0.5 \times 7.466 = 3.733$$

$$\text{normalised error, } E_n = \frac{2.39}{3.733} = 0.641$$

3. **Mann-Whitney U test.** In the effort of comparing the performance of GA and PSO, Tim has implemented both GA and PSO to solve the same problem. Both algorithms have been executed for six times each, and the solutions are recorded in Table 3.

Table 3: Solutions provided for the six runs by GA and PSO

Run	1	2	3	4	5	6
GA	-0.031	0.394	0.407	0.389	0.375	0.246
PSO	-0.416	0.263	0.376	0.279	0.042	-0.247

The fitness of the solution can be calculated using Equation 1.

$$f(x) = x^2 + 2x + 2 \quad (1)$$

Using Mann-Whitney U test, evaluate if any of the algorithms perform better than the other. Note that the higher fitness corresponds to better chromosome.

Solution:

Calculate fitness.

Run	1	2	3	4	5	6
GA	-0.031	0.394	0.407	0.389	0.375	0.246
Fitness of GA	1.939	2.943	2.980	2.929	2.891	2.553
PSO	-0.416	0.263	0.376	0.279	0.042	-0.247
Fitness of PSO	1.341	2.595	2.893	2.636	2.086	1.567

Arrange fitness from best (highest to lowest) to worst.

GA	GA	GA	PSO	GA	PSO	PSO	GA	PSO	GA	PSO	PSO
2.980	2.943	2.929	2.893	2.891	2.636	2.595	2.553	2.086	1.939	1.567	1.341

	1 st GA	2 nd GA	3 rd GA	4 th GA	5 th GA	6 th GA	U_1
# of PSO better	0	0	0	1	3	4	8

	1 st PSO	2 nd PSO	3 rd PSO	4 th PSO	5 th PSO	6 th PSO	U_2
# of GA better	3	4	4	5	6	6	28

$$U = \min(U_1, U_2) = \min(8, 28) = 8$$

The critical value for $n_1 = 6$ and $n_2 = 6$ is 7.

$U = 8 > 7$, thus the null hypothesis is not rejected. Statistically the two algorithms do not provide significantly different outcome.

4. **Mann-Whitney U test.** Two evolutionary algorithms are designed to determine the best configuration of a neural network. Daniel intends to compare the performance of the two evolutionary algorithms through the accuracies of the neural network configurations provided.

EA₁ produces neural networks with accuracies 0.95, 0.89, 0.85, 0.80, and 0.88 whereas EA₂ produces neural networks with accuracies 0.70, 0.93, 0.90, 0.98, 0.82, 0.75, and 0.86.

Using Mann-Whitney U test to compare the performance of the two evolutionary algorithms.

Solution:

Arrange fitness from best (highest to lowest) to worst.

EA ₂	EA ₁	EA ₂	EA ₂	EA ₁	EA ₁	EA ₂	EA ₁	EA ₂	EA ₁	EA ₂	EA ₂
0.98	0.95	0.93	0.9	0.89	0.88	0.86	0.85	0.82	0.8	0.75	0.7

	1 st EA ₁	2 nd EA ₁	3 rd EA ₁	4 th EA ₁	5 th EA ₁	U ₁
# of EA ₂ better	1	3	3	4	5	16

	1 st EA ₂	2 nd EA ₂	3 rd EA ₂	4 th EA ₂	5 th EA ₂	6 th EA ₂	7 th EA ₂	U ₂
# of EA ₁ better	0	1	1	3	4	5	5	19

$$U = \min(U_1, U_2) = \min(16, 19) = 16$$

The critical value for $n_1 = 5$ and $n_2 = 7$ is 6.

$U = 16 > 6$, thus the null hypothesis is not rejected. Statistically the performance of the two configurations are not significantly different.

5. **Confusion Matrices.** Construct the confusion matrix for the data in Table 4.

Table 4: Expected and predicted data

Index	Predicted Class	Expected Class	Index	Predicted Class	Expected Class
1	A	A	26	A	B
2	A	A	27	C	B
3	B	A	28	B	B
4	A	A	29	C	B
5	A	A	30	C	B
6	A	A	31	B	B
7	A	A	32	A	B
8	C	A	33	B	B
9	A	A	34	B	B
10	A	A	35	C	C
11	A	A	36	C	C
12	B	A	37	A	C
13	A	A	38	B	C
14	A	A	39	A	C
15	A	A	40	C	C
16	A	A	41	A	C
17	C	A	42	C	C
18	B	B	43	B	C
19	B	B	44	A	C
20	B	B	45	C	C
21	B	B	46	C	C
22	B	B	47	A	C
23	B	B	48	C	C
24	B	B	49	C	C
25	B	B	50	C	C

		Predicted		
		A	B	C
Actual/Expected	A	13	2	2
	B	2	12	3
	C	5	2	9

- (a) Calculate, for the system,
i. the accuracy,

Solution:

$$\text{accuracy} = \frac{13 + 12 + 9}{50} = \frac{34}{50} = 0.68 \text{ or } 68\%$$

- ii. the precision of predicting each classes, and

Solution:

$$\begin{aligned}\text{precision} &= \frac{TP}{TP + FP} \\ \text{precision for A} &= \frac{13}{13 + 2 + 5} = \frac{13}{20} = 0.65 \text{ or } 65\% \\ \text{precision for B} &= \frac{12}{12 + 2 + 2} = \frac{12}{16} = 0.75 \text{ or } 75\% \\ \text{precision for C} &= \frac{9}{9 + 3 + 2} = \frac{9}{14} = 0.6429 \text{ or } 64.29\%\end{aligned}$$

- iii. the recall of predicting each classes.

Solution:

$$\begin{aligned}\text{recall} &= \frac{TP}{TP + FN} \\ \text{recall for A} &= \frac{13}{13 + 2 + 2} = \frac{13}{17} = 0.7647 \text{ or } 76.47\% \\ \text{recall for B} &= \frac{12}{12 + 2 + 3} = \frac{12}{17} = 0.7059 \text{ or } 70.59\% \\ \text{recall for C} &= \frac{9}{9 + 5 + 2} = \frac{9}{16} = 0.5625 \text{ or } 56.25\%\end{aligned}$$

6. **Confusion Matrices.** An artificial neural network is designed to recognise between three types of vehicles (car, van, bus) from the footage of a CCTV device. The neural network has been tested on 1000 images with 400 images of cars and 300 images of bus.

The system accuracy is recorded to be 90%. The precision for recognition of van is 0.875. The recalls for the recognition of car and bus are 0.875 and 0.9 respectively. There are 365 images have been recognised as car, and 320 images as van. 5% of the images that contain van are recognised as bus. Construct the confusion matrix for the neural network.

Solution:

We first construct the confusion matrix with unknowns.

		Predicted			
		car	van	bus	
Actual	car	A	B	C	= 400
	van	D	E	F	
	bus	G	H	I	= 300
					1000

$$D + E + F = 1000 - 400 - 300 = 300$$

$$\text{system accuracy} = 0.9$$

$$\frac{A + E + I}{1000} = 0.9$$

$$A + E + I = 900 \quad (1)$$

$$\text{precision of van} = 0.875$$

$$\frac{E}{B + E + H} = 0.875$$

$$\text{recall of car} = 0.875$$

$$\frac{A}{A + B + C} = 0.875$$

$$\frac{A}{400} = 0.875$$

$$A = 350$$

$$\text{recall of bus} = 0.9$$

$$\frac{I}{G + H + I} = 0.9$$

$$\frac{I}{300} = 0.9$$

$$I = 270$$

$$\text{From (1)} \Rightarrow 350 + E + 270 = 900$$

$$E = 280$$

$$A + D + G = 365$$

$$B + E + H = 320$$

		Predicted			
		car	van	bus	
Actual	car	A	B	C	= 400
	van	D	E	F	
	bus	G	H	I	= 300
					1000
		365	320		

$$\frac{F}{D + E + F} = 0.05$$

$$\frac{F}{300} = 0.05$$

$$F = 15$$

$$C + F + I = 1000 - 365 - 320 = 315$$

$$C = 315 - 15 - 270 = 30$$

$$B = 400 - 30 - 350 = 20$$

$$D = 300 - 280 - 15 = 5$$

$$G = 365 - 350 - 5 = 10$$

$$H = 300 - 10 - 270 = 20$$

		Predicted		
		car	van	bus
Actual	car	350	20	30
	van	5	280	15
	bus	10	20	270

7. **Chi-Square test.** A decision tree is used to classify 50 data points into 5 classes, A, B, C, D, and E. The data set contains 10 data points for each class. The result provided by the decision tree classified 5 points to class A, 8 points to B, 15 points to C, 12 points to D, and the rest to E. Use chi-square test, with 0.05 level of significance, to comment on the performance of the decision tree.

Solution:

Class	A	B	C	D	E
Actual, E	10	10	10	10	10
Predicted, O	5	8	15	12	10
$O - E$	-5	-2	5	2	0
$(O - E)^2$	25	4	25	4	0
$\frac{(O-E)^2}{E}$	2.5	0.4	2.5	0.4	0

$$\chi^2 = 5.8$$

Degree of freedom, $df = 5 - 1 = 4$

At 0.05 level of significance, the critical value is 9.488.

As $\chi^2 = 5.8 < 9.488$, we do not reject the null hypothesis. The difference between the actual distribution and the predicted distribution by the decision tree is not significant.

8. **Chi-Square test.** A neural network is trained to perform classification on the same data set as in question (7). The result of the neural network shows 3 points for class A, 12 points for B, 5 points for C, 18 points for D, and the rest for E.

Compare the performance of the neural network and the decision tree from (7) using chi-square test at 0.05 level of significance.

Solution:

Class	A	B	C	D	E
Actual, E	10	10	10	10	10
Predicted, O	3	12	5	18	12
$O - E$	-7	2	-5	8	2
$(O - E)^2$	49	4	25	64	4
$\frac{(O-E)^2}{E}$	4.9	0.4	2.5	6.4	0.4

$$\chi^2 = 14.6$$

As $\chi^2 = 14.6 > 9.488$, the null hypothesis is rejected. The difference between the actual distribution and the predicted distribution by the neural network is significant.

Therefore the decision tree performs better than the neural network.